

EC3214 — International Finance

Summer 2024 Examination · Paper 1 · Full Marking Scheme

Module	EC3214 — International Finance
Internal Examiner	Dr Zhengnan Yin
Institution	University College Cork
Session	Summer 2024, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculators only

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. Where a question part carries a numerical result forward from an earlier step, an answer that applies the correct method consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Source basis. A1(c)'s PPP extensions and A3(a)'s account of the 2007-2009 financial crisis both follow the module's own assigned readings closely (including the BIS working paper on the 2007-08 financial turmoil and its "archetypal boom-bust" framing). A2(b)'s treatment of how domestic interest rates affect the exchange rate in the flexible- price monetary model, and A2(c)'s discussion of the Meese-Rogoff finding, follow the standard international finance literature (Frenkel, 1976; Meese and Rogoff, 1983) since these weren't set out in full in the uploaded material — flagged at the relevant questions.

Section A · Question 1

Triangulation, PPP and the real exchange rate, and PPP's extensions — 100 marks

A1(a)

[40 marks]

EUR/USD and AUD/USD are traded as standard interbank currency pairs and are showing the following prices: EUR/USD Bid 1.2542, Offer 1.2568; AUD/USD Bid 0.6276, Offer 0.6298. However, a customer would like a price in EUR/AUD. Using triangulation, calculate the no-arbitrage bid-offer quote for EUR/AUD.

A1(b)

[30 marks]

Briefly explain the relationship between Purchasing Power Parity (PPP) and the Real Exchange Rate.

A1(c)

[30 marks]

Based on your readings, discuss some extensions of PPP theory to account for observed variability in real exchange rates across countries and through time.

FULL MODEL ANSWER

Setting up the triangulation: both quoted pairs share USD as the common currency, so the EUR/AUD cross rate can be built by combining the two USD-denominated quotes such that the USD terms cancel. The standard cross-triangulation rule for bid and offer separately is:

Formula: $\text{EUR/AUD bid} = (\text{EUR/USD bid}) \div (\text{AUD/USD offer})$ $\text{EUR/AUD offer} = (\text{EUR/USD offer}) \div (\text{AUD/USD bid})$

Why bid pairs with offer (not bid with bid): the dealer's **bid** for EUR/AUD is the rate at which the dealer buys EUR from a customer, paying in AUD. To do this, the dealer synthetically buys EUR with USD (at the EUR/USD *bid*, since the dealer is buying EUR) and then must acquire that USD by buying it with AUD — which means selling AUD, at the AUD/USD *offer* (the worse rate for whoever is selling AUD, which protects the dealer's margin). The dealer's **offer** works the same way in reverse, using the EUR/USD offer and the AUD/USD bid.

Quote	Calculation	Result
EUR/AUD bid	$1.2542 \div 0.6298$	1.9914
EUR/AUD offer	$1.2568 \div 0.6276$	2.0025

Result: the no-arbitrage EUR/AUD quote is **1.9914 / 2.0025** — meaning the dealer will buy EUR (paying out AUD) at 1.9914 AUD per EUR, and sell EUR (receiving AUD) at 2.0025 AUD per EUR. The offer exceeds the bid, as it always must for a coherent two-way price, giving the dealer a positive spread on the synthetic cross exactly as they'd have on either underlying pair.

EXAM TECHNIQUE

A common error is dividing bid by bid or offer by offer, which produces an internally inconsistent quote (sometimes even offer < bid) — always match bid with offer and offer with bid across the two legs, since each synthetic transaction genuinely does route through two separate trades, one at each pair's less favourable side from the dealer's perspective.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly identifies that USD is the common currency to cancel AND correctly applies bid-with-offer, offer-with-bid triangulation AND arrives at 1.9914 / 2.0025 (or a consistent own-figure result) AND the offer exceeds the bid as required.
21-30	21-30	Correct triangulation method, but an arithmetic error in the final figures.
11-20	11-20	Incorrect method (e.g. bid-with-bid) but carried through consistently, or only one of the two (bid/offer) correctly derived.
0-10	0-10	No coherent triangulation attempted.

The relationship: the real exchange rate is defined as $Q = SP^*/P$ — the nominal exchange rate S adjusted for relative price levels. Absolute PPP ($P = SP^*$) holds exactly if and only if **$Q = 1$** : by direct substitution, $Q = SP^*/P = SP^*/(SP^*) = 1$. So PPP and the real exchange rate are two ways of describing the same underlying claim — PPP asserts that the real exchange rate is constant and equal to 1, meaning a given basket of goods costs the same everywhere once converted to a common currency, leaving no genuine difference in purchasing power once the nominal exchange rate is accounted for.

Why deviations of Q from 1 matter economically: when $Q \neq 1$, the real exchange rate is capturing genuine differences in relative competitiveness — a country whose real exchange rate has appreciated (Q risen, in a convention where higher Q means the country's goods are relatively more expensive once converted to a common currency) has become less price-competitive internationally, regardless of what its nominal exchange rate has done, since nominal exchange-rate movements can be offset or reinforced by relative inflation differences.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly states the real exchange rate definition ($Q=SP^*/P$) AND correctly explains that PPP holding is equivalent to $Q=1$ AND explains the economic meaning of Q deviating from 1 (competitiveness).
15-22	15-22	Correctly states $Q=SP^*/P$ and that PPP implies $Q=1$, without further economic interpretation.
8-14	8-14	Vague reference to a relationship between PPP and the real exchange rate without the $Q=SP^*/P$ definition or the $Q=1$ result.
0-7	0-7	No credible answer given.

Why observed real exchange rates deviate from PPP's $Q=1$

prediction: the literature has identified several distinct extensions and refinements that help explain persistent, observed variability in real exchange rates, beyond simple measurement or short-run noise.

- **The Balassa-Samuelson effect:** productivity growth is typically faster in the tradable-goods sector than the non-tradable (services) sector, and faster in richer, faster-growing economies. As tradable-sector productivity rises, wages are bid up economy-wide (including in the non-tradable sector, where productivity hasn't risen correspondingly), pushing up the relative price of non-tradables and hence the overall price level — meaning faster-growing economies systematically have persistently higher real exchange rates (their currencies look "overvalued" relative to naive PPP), not because of any market failure but because of structural productivity differences.
- **Non-traded goods more generally:** even without the specific productivity-growth mechanism above, any basket that includes non-traded services (haircuts, restaurant meals, local housing) will show real exchange rate variability, since the law of one price simply cannot operate on goods that structurally cannot be arbitrated across borders (see the EC3214 Summer 2025 scheme, Q. A1(a), for a direct numerical illustration).
- **Transport costs and trade barriers:** tariffs, transport costs, and other trade frictions create a genuine band within which prices can differ across countries without triggering arbitrage — the price gap has to be large enough to cover these costs before arbitrage becomes profitable, so real exchange rates can wander within that band without violating the (frictional) law of one price.
- **Sticky prices and slow adjustment (Dornbusch-style):** even where PPP would eventually hold as a long-run equilibrium, nominal prices adjust slowly while the nominal exchange rate can move quickly, so the real exchange rate can display substantial short- and medium-run variability purely from the different speeds at which nominal exchange rates and nominal prices adjust (see Q. A3(b)/(c) on the Dornbusch overshooting model).
- **Demand-side and portfolio factors:** shifts in risk premia, safe-haven demand, and international portfolio rebalancing can move nominal (and hence real) exchange rates for reasons that have nothing to do with relative goods prices at all — a channel entirely outside the goods-market arbitrage logic PPP is built on.

EXAM TECHNIQUE

A strong answer doesn't just list these — it explains *why* each one specifically breaks the law-of-one-price-to-PPP chain (see Q. A1(b)): Balassa-Samuelson and non-traded goods break the "same basket, identically weighted" assumption; transport costs/tariffs break the "frictionless arbitrage" assumption; sticky prices break the "instantaneous adjustment" assumption; portfolio factors introduce forces entirely outside the goods-market channel PPP describes.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	At least three distinct extensions/explanations correctly given (Balassa-Samuelson, non-traded goods, transport costs/tariffs, sticky prices, portfolio/risk factors) AND each is explained in terms of which specific PPP assumption it breaks, not just named.
15-22	15-22	At least two extensions correctly given and explained.
8-14	8-14	Only one extension correctly explained, or several named without explanation of the underlying mechanism.
0-7	0-7	No credible extensions given.

Section A · Question 2

The flexible-price monetary model: assumptions, comparative statics, and an evaluation — 100 marks

A2(a)

[30 marks]

What are the assumptions underlying the flexible price monetary model of a floating exchange rate?

A2(b)

[40 marks]

Explain, under a floating exchange rate regime, using the flexible price monetary model how each of the following will affect the home country exchange rate: an expansion in the home country money supply; an increase in national income in the home country; an increase in the foreign price level; an increase in domestic interest rates. Support your answer with an appropriate diagram.

A2(c)

[30 marks]

Using empirical and theoretical evidence, evaluate the usefulness of this approach.

FULL MODEL ANSWER

- **Purchasing power parity holds continuously:** absolute PPP ($S = P/P^*$) is assumed to hold at every point in time, not just in the long run — the exchange rate is always exactly consistent with relative price levels.
- **Prices are fully flexible:** both domestic and foreign price levels adjust instantaneously to clear their respective money markets, with no stickiness of any kind.
- **Output is fixed at its full-employment level:** real income y is determined independently of monetary factors (by the economy's real productive capacity), so monetary shocks cannot move output — **money is neutral**.
- **Stable money demand function:** real money demand is a stable, well-behaved function of income (and, in extended versions, the interest rate), so that $M_o^s = kPy$ (or its interest-rate-augmented version) reliably describes money market equilibrium.
- **Perfect capital mobility / uncovered interest parity:** financial markets are assumed to be perfectly integrated internationally, so that interest rate

differentials reflect expected exchange rate movements (see the EC3214 Summer 2025 scheme, Q. A1(c), for the UIP condition itself).

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	At least four of the five core assumptions correctly stated and explained (continuous PPP, full price flexibility, fixed/full-employment output, stable money demand, perfect capital mobility/UIP).
15-22	15-22	At least three assumptions correctly stated.
8-14	8-14	Only one or two assumptions correctly stated, or stated without explanation.
0-7	0-7	No credible assumptions given.

Setting up the model: money market equilibrium gives $P = M_o^s/(ky)$; PPP gives $S = P/P^*$ (domestic currency units per unit of foreign currency — under this convention, a **rise** in S is a domestic currency **depreciation**, and a **fall** is an **appreciation**).

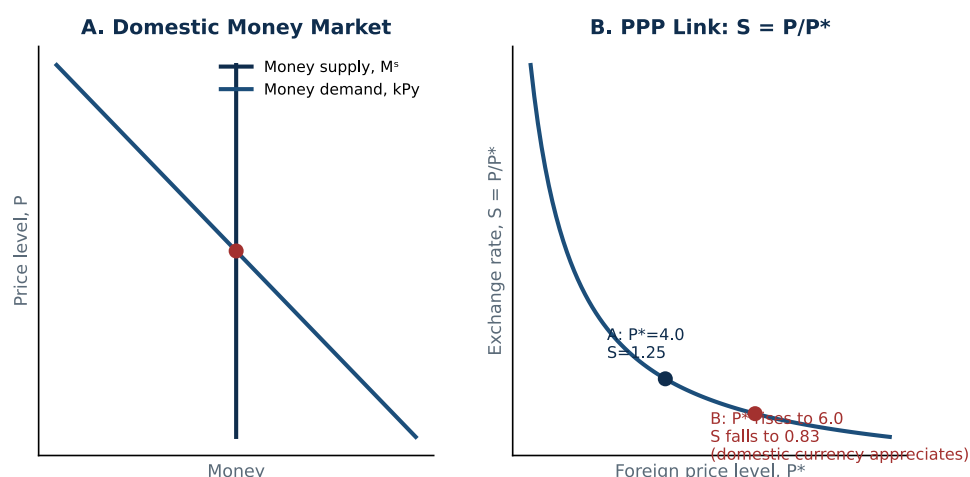


Fig. 1 — The domestic money market determines P (Panel A); the PPP link then maps P into S given P^* (Panel B). Each of the four shocks below works by shifting one of these two relationships.

Shock	Effect on P	Effect on S	Direction
Home money supply expansion ($M_o^s \uparrow$)	P rises proportionally ($P = M_o^s/(ky)$)	S rises proportionally ($S = P/P^*$)	Depreciation
Home national income increase ($y \uparrow$)	P falls (money demand ky rises, M_o^s fixed)	S falls	Appreciation
Foreign price level increase ($P^* \uparrow$)	No change (domestic side unaffected)	S falls ($S = P/P^*$, P^* in denominator)	Appreciation
Domestic interest rate increase ($i \uparrow$)	P rises (real money demand falls as i rises, so P must rise to keep M_o^s/P at the lower required real balance)	S rises	Depreciation

The first three, briefly: a money supply expansion raises P proportionally (classic quantity-theory logic) and, via PPP, depreciates the currency by the same proportion. A rise in national income raises real money *demand*; with the money *supply* unchanged, equilibrium requires P to fall (domestic deflation),

which appreciates the currency via PPP. A rise in the foreign price level, with nothing on the domestic side changing, directly appreciates the domestic currency through the PPP identity alone (see the EC3214 Summer 2025 scheme, Q. A2(b), for the same result derived in full).

The fourth — domestic interest rates — deserves special care, since the result is genuinely counterintuitive: in the extended flexible-price monetary model with an interest-elastic money demand function ($M_d/P = kY^a i^{-b}$, where money demand falls as the interest rate i rises — the opportunity cost of holding non-interest-bearing money is higher), a **rise** in the domestic interest rate **reduces** real money demand. For the money market to stay in equilibrium with an unchanged money supply, the price level P must **rise** to bring real money balances (M_o^s/P) down to match the now-lower real money demand — and a higher P , via PPP, means the domestic currency **depreciates**. This is the opposite of the "higher rates attract capital, currency appreciates" intuition many people bring to this question, and it is exactly right within this model: the underlying logic is that a higher domestic *nominal* interest rate typically reflects higher *expected inflation* (via the Fisher effect), and if the currency is expected to depreciate more in the future because of that higher expected inflation, uncovered interest parity requires it to **depreciate immediately today** too, not appreciate.

EXAM TECHNIQUE

This fourth result is the single most commonly mishandled part of this question — many answers default to the simpler "higher rates → capital inflow → appreciation" story, which is the Mundell-Fleming/portfolio-balance logic, not the flexible-price monetary model's own prediction. Within *this specific model*, higher domestic nominal rates (driven by higher expected inflation) depreciate, not appreciate, the currency — get this backwards and the direction in the mark band below will be marked as incorrect even if the reasoning for the other three shocks is otherwise strong.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	All four comparative statics correctly signed and explained via the $P=M_o^s/(ky)$ and $S=P/P^*$ mechanism AND the domestic interest rate result specifically correctly identified as depreciation (not the more intuitive appreciation), with the Fisher-effect/expected-inflation reasoning given AND the diagram correctly illustrates the mechanism.
21-30	21-30	At least three of the four correctly signed and explained, with the interest-rate case being the most likely to be missed or reversed.
11-20	11-20	Only two of the four correctly explained.
0-10	0-10	Fewer than two correctly explained, or no credible mechanism given for any of them.

Theoretical weaknesses: the model's cleanest predictions rest on assumptions that are individually questionable: PPP is assumed to hold continuously, but (see Q. A1(c)) empirically it holds poorly in the short and medium run; output is assumed fixed at full employment, ruling out by assumption the very real-output effects of monetary policy that Mundell-Fleming was built to capture; and money demand is assumed stable, which has itself been empirically unstable at various points (a criticism that also applies within closed-economy monetary economics).

The interest-rate prediction is a particular empirical embarrassment: the model's prediction that higher domestic nominal rates

depreciate the currency (Part (b)) sits awkwardly next to the well-documented **forward premium puzzle** (Fama, 1984) — the empirical finding that high-interest-rate currencies tend, on average, to *appreciate* rather than depreciate as uncovered interest parity would predict, the opposite of what the model's clean UIP-based logic implies should happen on average.

The Meese-Rogoff result — the most damaging evidence of all: Meese and Rogoff (1983) famously found that monetary exchange-rate models, including the flexible-price version examined here, **fail to out-forecast a naive random walk** in out-of-sample exchange rate prediction — even when the models are fed the *actual, realised* values of their own explanatory variables (money supplies, output, etc.) after the fact, rather than forecasts of them. This is a striking result: a model that cannot beat "tomorrow's rate will be about the same as today's rate" even with perfect hindsight on its own fundamentals casts serious doubt on the model's short- to medium-run empirical usefulness, and the puzzle has never been fully resolved in the decades since.

Where the model does better: longer-run, lower-frequency evidence is considerably kinder to the monetary approach — several studies find that monetary fundamentals (relative money supplies, relative output) do have genuine explanatory power for exchange rates over multi-year horizons, and extreme cases (hyperinflations, very large sustained money-growth differentials) show a strong, clearly monetary-driven relationship between money growth and currency depreciation, consistent with the model's core quantity-theory logic.

Theoretically clean but empirically weak in the short-to-medium run; more useful as a long-run, qualitative framework than as a forecasting tool.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	At least two specific theoretical weaknesses correctly identified (PPP failure, fixed-output assumption) AND the Meese-Rogoff finding correctly cited and explained AND the model's better long-run performance is acknowledged, giving a balanced, evidence-based verdict.
15-22	15-22	At least one specific empirical critique given (Meese-Rogoff or the forward premium puzzle), with a verdict, but the balanced long-run counterpoint is missing.
8-14	8-14	Generic statement that "the model doesn't always work" without citing specific empirical evidence.
0-7	0-7	No credible evaluation given.

Question A2 Total: 100 Marks

Section A · Question 3

The 2007-2009 financial crisis, the Dornbusch model, and testing for overshooting — 100 marks

A3(a)

[30 marks]

Explain the source of the global financial crisis of 2007-2009.

A3(b)

[30 marks]

Using the Dornbusch model, consider the predicted effects of a monetary expansion under a flexible exchange rate regime. Support your answer with an appropriate diagram.

A3(c)

[40 marks]

The exchange rate of the British Pound against the dollar since the global financial crisis has followed this path: Average for month, GBP per US\$: Mar-08 0.4995; Mar-09 0.70607; Dec-14 0.63779. Is this evidence of overshooting? If yes, is it an indication that the Dornbusch model may be correct — i.e. a good explanation of how exchange rates behave? If not, why not?

FULL MODEL ANSWER

The trigger: US subprime mortgages. the proximate trigger was rising delinquencies in the US subprime mortgage market — loans extended to borrowers with weak credit histories, often on the assumption that continually rising house prices would let them refinance before payments reset higher. When US house price growth slowed and reversed, subprime delinquencies rose sharply starting in 2006-07, and interbank markets in the US and euro area came under severe, unexpected strain in August 2007.

The deeper cause: a boom-bust cycle of financial imbalance. the module's own assigned BIS reading frames the crisis as "the archetypal example of financial instability... that follows the build-up of financial imbalances in good times" — the subprime market was where the imbalance happened to be concentrated, but the underlying dynamic (a prolonged period of generalised, aggressive risk-taking during a boom, masked by rising asset prices and strong growth) was the deeper, more general cause. Financial crashes, on this reading, are not random bad-apple failures — they systematically follow booms, precisely because almost every institution looks safe and well-capitalised while a boom is still inflating asset values.

The amplification mechanisms

- **Securitisation and the originate-to-distribute model:** subprime mortgages were pooled and repackaged into mortgage-backed securities and collateralised debt obligations, sold on to investors worldwide — spreading the eventual losses far beyond the originating lenders, but also spreading the *uncertainty* about where losses actually sat once the underlying loans began defaulting, which is exactly what froze the interbank market in August 2007 (no bank could confidently assess its counterparties' exposure).
- **Credit rating failures:** many of these securitised products carried investment-

grade ratings that turned out not to reflect their true underlying risk, encouraging wider holding of the assets — including by institutions with mandates restricting them to highly-rated instruments — than the true risk profile would have justified.

- **Leverage and a "reintermediation" wave:** banks had moved much of this exposure off their own balance sheets into structured vehicles, but as the crisis unfolded they were forced to bring much of it back onto their balance sheets (an unanticipated "reintermediation"), straining their own capital precisely when capital was hardest to raise.
- **Global interconnectedness:** the securitised products, and the banks exposed to them, were held internationally, which is why a US-originated shock produced interbank stress simultaneously in the US, euro area, UK, Switzerland, Canada, Australia and Japan — culminating, after Lehman Brothers' collapse in September 2008, in a genuinely global financial and economic crisis.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies the subprime mortgage trigger AND correctly frames the deeper cause as a boom-bust financial-imbalance cycle rather than a purely subprime-specific failure AND at least two amplification mechanisms (securitisation, rating failures, reintermediation, global interconnectedness) correctly explained.
15-22	15-22	Correctly identifies subprime mortgages as the trigger with at least one amplification mechanism explained, but doesn't distinguish the trigger from the deeper systemic cause.
8-14	8-14	Vague reference to "the housing market" or "banks taking too much risk" without specific mechanisms.
0-7	0-7	No credible account of the crisis given.

Setting up the Dornbusch model: Dornbusch (1976) modifies the flexible-price monetary model (Q. A2) by assuming **domestic prices are sticky in the short run but fully flexible in the long run** — a hybrid that nests Mundell-Fleming-style short-run rigidity with the Monetary Model's long-run flexible-price outcome.

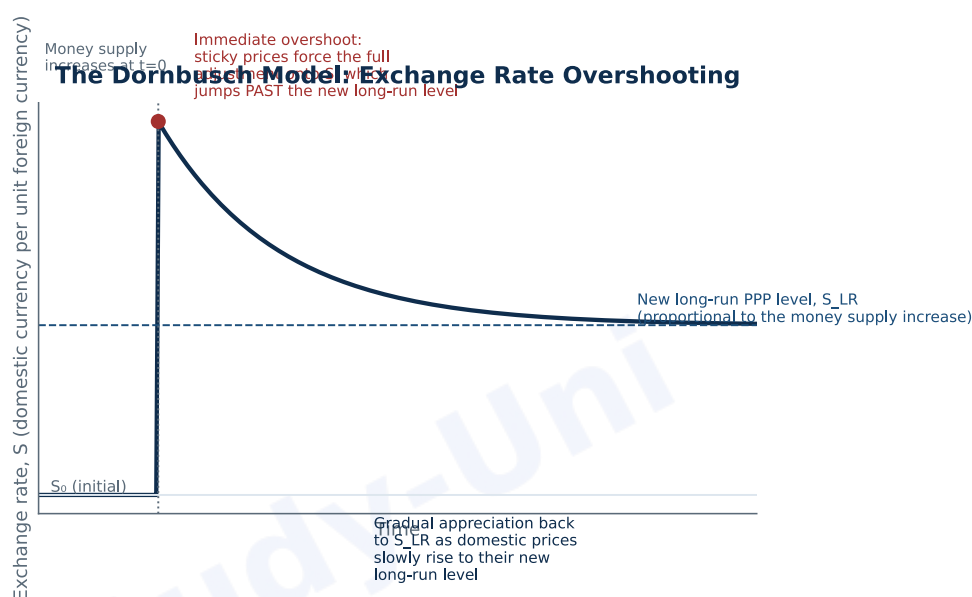


Fig. 1 — Following a permanent, unanticipated money supply increase, the exchange rate jumps immediately past its new long-run level (overshooting) because sticky prices force the entire short-run adjustment onto S , then gradually appreciates back to the new, higher long-run PPP level as domestic prices slowly catch up.

The mechanism, step by step: with domestic prices fixed in the short run, an increase in the nominal money supply is, for a moment, an increase in the

real money supply too (since P hasn't moved yet) — this lowers the domestic interest rate (standard liquidity-effect logic). With capital highly mobile, the lower domestic interest rate immediately triggers a capital **outflow**, which causes the domestic currency to **depreciate sharply and immediately**. Crucially, because uncovered interest parity must still hold even in the short run, the currency has to be expected to *appreciate* thereafter (to compensate investors for the now-lower domestic interest rate) — meaning the initial depreciation must **overshoot** the new long-run equilibrium level, so that the subsequent gradual appreciation back toward it is exactly what makes holding the (temporarily lower-yielding) domestic currency worthwhile in the interim.

The long-run outcome: as domestic prices gradually rise to their new, higher long-run level (consistent with the pure Monetary Model's prediction — see Q. A2(b)), the real money supply falls back toward its original level, the domestic interest rate rises back to its original level, and the exchange rate appreciates back from its overshoot low toward the new long-run PPP-consistent level — which is itself proportionally more depreciated than the original rate, by the same percentage as the money supply increase, exactly matching the pure Monetary Model's long-run prediction.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly states the sticky-price-short-run/flexible-price-long-run assumption AND correctly traces the full mechanism (money supply up → interest rate down → capital outflow → immediate overshoot depreciation → gradual price adjustment → gradual appreciation back to the new long-run level) AND correctly connects the long-run outcome to the pure Monetary Model's prediction.
15-22	15-22	Correctly identifies that overshooting occurs and diagrams it, but the UIP-based reasoning for <i>why</i> it must overshoot (rather than just moving directly to the new long-run level) is missing.
8-14	8-14	General description of "exchange rate overshooting" without the specific sticky-price/UIP mechanism.
0-7	0-7	No credible mechanism given.

The data pattern: GBP fell sharply against the dollar from Mar-08 to Mar-09 (0.4995 to 0.70607 GBP per \$ — a **41.4% depreciation** of sterling against the dollar over that year), then partially reversed by Dec-14 (down to 0.63779 GBP per \$ — a **9.7% appreciation** of sterling from its Mar-09 trough), leaving GBP still **27.7% weaker** against the dollar than it had been at the Mar-08 starting point.

Is this evidence of overshooting? the *shape* of the pattern — a sharp initial move followed by a partial reversal in the same direction as a longer-run adjustment — is superficially consistent with the qualitative pattern the Dornbusch model predicts (Part (b)'s diagram). In that narrow sense, **yes**, this is consistent with overshooting-type dynamics.

But this is NOT strong evidence that the Dornbusch model specifically is correct.

Why the pattern doesn't validate the Dornbusch mechanism

specifically: the Dornbusch model's overshooting prediction is the response to a **specific, clean, unanticipated monetary policy shock** under sticky prices — it is not a general theory that predicts "any large exchange rate move followed by a partial reversal is overshooting." The 2008-09 period was nothing

like a clean, isolated monetary shock:

- **Multiple, simultaneous shocks:** the 2008-09 window saw a systemic global financial crisis (Q. A3(a)), not an isolated UK monetary policy action — a severe **flight-to-safety** toward the US dollar (the world's principal safe-haven currency during a global panic) can easily explain a sharp GBP depreciation with nothing to do with UK monetary policy or sticky domestic prices at all.
- **Both central banks were acting simultaneously:** the Bank of England and the Federal Reserve both cut interest rates sharply and moved to unconventional policy (quantitative easing) over this period — Dornbusch's clean mechanism assumes a shock to one country's money supply with the other held fixed, not a global, near-simultaneous policy response on both sides.
- **Risk premia, not just interest differentials:** UIP itself (the mechanism that forces overshooting in the Dornbusch model) is well known empirically to be unreliable, especially during periods of financial stress when risk premia move sharply and unpredictably (see Q. A2(c)'s discussion of the forward premium puzzle) — exactly the environment 2008-09 represents.
- **The timescale and incomplete reversal:** Dornbusch overshooting is meant to resolve as prices adjust over a period of months to a few years; here, GBP was still 27.7% weaker than its pre-crisis level nearly *seven years* later, in Dec-14 — consistent just as easily with a genuine, permanent structural shift in the UK's relative competitiveness or growth outlook (e.g. the UK financial sector's outsized exposure to the crisis) as with a slow-motion Dornbusch-style price adjustment still working its way through.

The bottom line: correlation with the *qualitative shape* of the overshooting pattern is not evidence that the Dornbusch *mechanism* — a specific causal story about sticky prices, interest rates, and UIP following a monetary shock — is what actually generated this particular currency move. Many alternative, at least equally plausible explanations (safe-haven flows, simultaneous multi-country policy responses, a genuine structural shift, breakdown of UIP under stress) fit the same data equally well or better, given how eventful and multi-causal the 2008-2014 period was. This is a standard identification problem in empirical international finance: a model's qualitative prediction matching a real-world pattern is suggestive, but nowhere near sufficient, to confirm that model's specific mechanism was the actual cause.

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly calculates the percentage moves AND correctly identifies the pattern as superficially consistent with overshooting AND correctly argues this is NOT strong evidence the Dornbusch model specifically is correct, citing at least two specific alternative explanations (safe-haven flows, simultaneous policy actions, UIP breakdown under stress, structural shift) AND articulates the general identification problem (pattern-matching $\neq$ mechanism confirmation).
21-30	21-30	Correctly calculates the moves and identifies the pattern as overshooting-like, with at least one alternative explanation given for why this doesn't confirm Dornbusch specifically.
11-20	11-20	Correctly calculates the moves, but concludes (without critical engagement) that this simply confirms the Dornbusch model is correct.
0-10	0-10	Percentage moves not correctly calculated, or no credible discussion given.

Question A3 Total: 100 Marks

